

Thermodynamic Dynamics for the TFPT Finite-Range Master-Hamiltonian Class

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Abstract

We isolate the uniformly finite-dimensional, finite-range Hamiltonian class actually assembled by the TFPT 2 + 1- and 3 + 1-dimensional probes: electric and plaquette gauge terms, gauge-dressed fermion hopping, the DET16 mirror projector, and the seam clock and charge terms. At the frozen 2 + 1-dimensional coefficients the interaction has range $R = 2$, largest local term $J = 12/5$, and interaction moment $\kappa = 16043/450$. A complete finite-range Lieb–Robinson argument gives

$$v_{\text{LR}} = 2e\kappa R = \frac{32086e}{225} = 387.639069990831.$$

We then prove in norm, rather than assume, that the finite-volume Heisenberg dynamics form a Cauchy net on every local observable and hence define a strongly continuous thermodynamic-limit automorphism group. Gauss constraints are handled on the gauge-invariant subalgebra of the ambient quasilocal algebra; naive embeddings of projected physical Hilbert spaces are neither needed nor asserted. Weak-* compactness gives at least one ground state and, for each positive inverse temperature, at least one KMS state. The numeric twin passes all eight checks and measures velocity 2.0, well below the proved class bound.

Contents

1	The concrete uniformly local class	1
1.1	Ambient tensor-product algebra	1
1.2	Uniform constants	2
2	The Lieb–Robinson theorem and explicit TFPT velocity	2
3	Norm thermodynamic limit	4
4	Gauss law and the correct consistent family	4
5	Existence of equilibrium states	5
6	Numeric twin	6
7	Combined theorem and exact boundary	6
	Conclusion	7

1 The concrete uniformly local class

1.1 Ambient tensor-product algebra

Let $\Gamma = \mathbb{Z}^2$ with graph distance d and coordination $z = 4$. For a finite $\Lambda \Subset \Gamma$, put

$$\mathcal{H}_\Lambda = \bigotimes_{x \in \Lambda} \mathcal{H}_x, \quad \mathcal{A}_\Lambda = \mathcal{B}(\mathcal{H}_\Lambda), \quad \iota_{\Lambda, \Lambda'}(A) = A \otimes \mathbf{1}_{\Lambda' \setminus \Lambda}. \quad (1.1)$$

One cell contains the four representative physical edge modes, a mirror Fock factor, the two positively oriented outgoing finite-group gauge links, and a four-state seam clock. Thus

$$\dim \mathcal{H}_x = 16 \cdot 2^{n_{\text{mir}}} \cdot 2^2 \cdot 4, \quad n_{\text{mir}} \in \{4, 16\}. \quad (1.2)$$

The DET4 numerical analogue has dimension 4096; the full DET16 cell has dimension 16 777 216. Only finiteness and a volume-independent upper bound are used below. The quasilocal algebra is

$$\mathcal{A} = \overline{\bigcup_{\Lambda \Subset \Gamma} \mathcal{A}_\Lambda}^{\|\cdot\|}. \quad (1.3)$$

Definition 1.1 (TFPT assembled interaction). *For finite $X \Subset \Gamma$, let $\Phi(X) = \Phi(X)^*$, with $\Phi(X) = 0$ unless X is one of the supports in Table 1. The free-boundary finite-volume Hamiltonian is*

$$H_\Lambda = \sum_{X \subseteq \Lambda} \Phi(X). \quad (1.4)$$

Scalar multiples of the identity may be removed termwise, since they do not change $e^{itH_\Lambda} A e^{-itH_\Lambda}$.

term	frozen operator and coefficient	$\ \Phi(X)\ $	$ X $	diam X
electric	$\frac{g^2}{2} E_\ell^2$, $g = 1.2$, with each positive link assigned to its tail cell	$18/25 = 0.72$	1	0
plaquette	$\frac{1}{2g^2}(2 - U_p - U_p^*)$; its dynamical representative is $-\frac{1}{2g^2}(U_p + U_p^*)$	$25/36$	4	2
hopping	$-t \sum_{a=1}^4 (c_{y,a}^* U_{xy} c_{x,a} + \text{h.c.})$, $t = 0.6$	$\leq 4t = 12/5$	2	1
DET16	$2(\mathbf{1} - P_{\phi,x})$, where $P_{\phi,x}$ is the rank-one Spin(10)-invariant determinant projector	2	1	0
seam clock	$-0.5(\Psi_x + \Psi_x^*)$, $\Psi_x^4 = \mathbf{1}$	1	1	0
seam charge	$0.2(\Psi_x + \Psi_x^*)(n_{Q,x} - 1/2)/2$	$1/10$	1	0
staggered mass	$0.2 \sum_{a=1}^4 (-1)^x (n_{x,a} - n_{x,a}^{\text{bg}})$	$\leq 4/5$	1	0

Table 1: Supports and norm bounds extracted from the frozen assembled models. The positive plaquette term itself has norm at most $2/g^2 = 25/18$; after deleting its scalar part, the norm relevant to dynamics and commutators is $1/g^2 = 25/36$.

The determinant identity checked in `verification/v1002_spin10_mirror_projector.py` gives $P_{\phi,x} = P_{\phi,x}^* = P_{\phi,x}^2$, so its row has norm exactly two. The clock and charge bounds follow from $\|\Psi\| = 1$, $\|(\Psi + \Psi^*)/2\| = 1$, and $\|n_Q - 1/2\| = 1/2$.

1.2 Uniform constants

Define the finite-range interaction moment

$$\kappa := \sup_{x \in \Gamma} \sum_{X \ni x} |X| \|\Phi(X)\|. \quad (1.5)$$

There are two assigned electric links, four plaquettes, and four bonds incident on a square-lattice cell. Consequently

$$\begin{aligned} J &:= \sup_X \|\Phi(X)\| = \frac{12}{5}, \quad R = 2, \\ \kappa &\leq 2 \frac{18}{25} + 4 \cdot 4 \frac{25}{36} + 4 \cdot 2 \frac{12}{5} + 2 + 1 + \frac{1}{10} + \frac{4}{5} \\ &= \frac{16043}{450} = 35.651111111111 \dots \end{aligned} \tag{1.6}$$

These bounds are independent of Λ . More generally, bounded degree, finite R , finite local dimension and finite κ are the class hypotheses; the displayed numbers are the frozen assembled $2 + 1$ -dimensional specialization.

2 The Lieb–Robinson theorem and explicit TFPT velocity

Theorem 2.1 (Finite-range Lieb–Robinson theorem, full form used here). *Let (Γ, d) be a graph, let every local Hilbert space be finite dimensional with a uniform dimension bound, and let a self-adjoint interaction satisfy*

$$\Phi(X) = 0 \quad \text{if } \text{diam } X > R, \quad \sup_x \sum_{X \ni x} |X| \|\Phi(X)\| = \kappa < \infty. \tag{2.1}$$

For every finite Λ , define $\tau_t^\Lambda(A) = e^{itH_\Lambda} A e^{-itH_\Lambda}$. If $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$, and $d(X, Y) > 0$, then, uniformly in $\Lambda \supset X \cup Y$,

$$\left\| [\tau_t^\Lambda(A), B] \right\| \leq 2 \|A\| \|B\| |X| \exp \left[-\frac{d(X, Y)}{R} + 2e\kappa|t| \right]. \tag{2.2}$$

Thus an admissible Lieb–Robinson velocity is

$$v_{\text{LR}} = 2e\kappa R. \tag{2.3}$$

The same assertion holds for even observables in a CAR algebra, because disjoint even local algebras commute.

Proof. This is the original Lieb–Robinson iteration in the finite-range normalization (2.1); it is equivalent to the exponential Nachtergaele–Sims form. We include the complete estimate. Differentiate the commutator after removing all Hamiltonian terms supported entirely away from its evolving support. Duhamel iteration gives

$$\left\| [\tau_t^\Lambda(A), B] \right\| \leq 2 \|A\| \|B\| \sum_{n \geq 1} \frac{(2|t|)^n}{n!} a_n(X, Y), \tag{2.4}$$

where $a_n(X, Y)$ is the sum of $\prod_{j=1}^n \|\Phi(Z_j)\|$ over chains with $Z_1 \cap X \neq \emptyset$, $Z_j \cap Z_{j-1} \neq \emptyset$, and $Z_n \cap Y \neq \emptyset$. Choosing one intersection site at each step and using (2.1) yields

$$a_n(X, Y) \leq |X| \kappa^n. \tag{2.5}$$

Every member of a chain has diameter at most R ; hence no such chain can join X to Y unless $nR \geq d(X, Y)$. Put $m = \lceil d(X, Y)/R \rceil$. For $n \geq m$, $\mathbf{1}_{\{n \geq m\}} \leq e^{n-m}$, and therefore

$$\sum_{n \geq m} \frac{(2\kappa|t|)^n}{n!} \leq e^{-m} \sum_{n \geq 0} \frac{(2e\kappa|t|)^n}{n!} \leq e^{-d(X, Y)/R + 2e\kappa|t|}. \tag{2.6}$$

Substitution in (2.4) proves (2.2). Rewriting the exponent as $-[d(X, Y) - 2e\kappa R|t|]/R$ proves (2.3). In the fermionic case all Hamiltonian terms and physical observables under consideration are even, so the initial disjoint-support commutator and every step of the iteration are ordinary commutators. $\square$

The theorem is the finite-range specialization of Lieb and Robinson [1] and of the exponential bounds of Nachtergaele and Sims [2, 3]. For the TFPT constants,

$$v_{\text{LR}} = 2e \frac{16043}{450} 2 = \frac{32086e}{225} = 387.639069990831. \quad (2.7)$$

This deliberately conservative class bound includes all four flavors, plaquettes, and onsite terms. It is not a fitted physical signal speed.

3 Norm thermodynamic limit

Theorem 3.1 (Thermodynamic-limit dynamics). *For every $A \in \mathcal{A}_{\text{loc}} := \bigcup_{\Lambda} \mathcal{A}_{\Lambda}$, the net*

$$\tau_t(A) := \lim_{\Lambda \uparrow \Gamma} \tau_t^{\Lambda}(A) \quad (3.1)$$

exists in norm, uniformly for t in compact real intervals and independently of the exhausting sequence. It extends uniquely to a strongly continuous one-parameter group of C^ -automorphisms of $\mathcal{A}$.*

Proof. We give the standard Bratteli–Robinson/Nachtergaele–Sims proof with an explicit Cauchy bound. Let $A \in \mathcal{A}_X$, and let $\Lambda \subset \Lambda'$ contain the closed b -neighbourhood of X . Write $V = H_{\Lambda'} - H_{\Lambda}$, with both Hamiltonians embedded in $\mathcal{A}_{\Lambda'}$. Differentiating $\tau_{t-s}^{\Lambda'}(\tau_s^{\Lambda}(A))$ gives

$$\tau_t^{\Lambda'}(A) - \tau_t^{\Lambda}(A) = i \int_0^t \tau_{t-s}^{\Lambda'}([V, \tau_s^{\Lambda}(A)]) ds. \quad (3.2)$$

Every term Z in V reaches outside Λ . Since $\text{diam } Z \leq R$, $d(X, Z) \geq \rho := \max(0, b - R)$. Apply (2.2) termwise. If $q = e^{-1/R}$, then the square-lattice shell bound

$$\sum_{y: d(X, y) \geq \rho} e^{-d(X, y)/R} \leq 4|X|q^{\rho} \left(\frac{\rho+1}{1-q} + \frac{q}{(1-q)^2} \right) =: |X|S_R(\rho) \quad (3.3)$$

follows by assigning each y to a nearest point of X and using at most $4(n+1)$ sites in shell n . Assign each interaction Z to a site $y \in Z$ nearest X . Because $\text{diam } Z \leq R$, $e^{-d(X, Z)/R} \leq e e^{-d(X, y)/R}$, while $\sum_{Z \ni y} \|\Phi(Z)\| \leq \kappa$. Equations (3.2)–(3.3) therefore give, for $t \geq 0$,

$$\begin{aligned} \left\| \tau_t^{\Lambda'}(A) - \tau_t^{\Lambda}(A) \right\| &\leq 2 \|A\| |X| e \kappa |X| S_R(\rho) \int_0^t e^{2e\kappa s} ds \\ &= \|A\| |X|^2 S_R(\rho) (e^{2e\kappa t} - 1). \end{aligned} \quad (3.4)$$

The same bound holds with $|t|$. For fixed T , its right-hand side tends to zero as $b \rightarrow \infty$, uniformly for $|t| \leq T$. Hence the finite-volume dynamics are a norm-Cauchy net on local observables; this also proves independence of the exhaustion.

Each τ_t^{Λ} is isometric, linear, multiplicative and star-preserving. Norm limits preserve these properties. Applying the construction to $-t$ shows that τ_{-t} is the inverse, and taking limits in $\tau_{s+t}^{\Lambda} = \tau_s^{\Lambda} \tau_t^{\Lambda}$ gives the group law. For local A , continuity at $t = 0$ follows first in a fixed sufficiently large volume and then from the compact-time uniform limit. Density of $\mathcal{A}_{\text{loc}}$ and isometry extend both τ_t and strong continuity to all of $\mathcal{A}$. This is precisely the conclusion of the standard thermodynamic-dynamics theorem, commonly cited as Bratteli–Robinson, Theorem 6.2.11 [4]. $\square$

4 Gauss law and the correct consistent family

Let $\mathcal{G}_{\text{loc}}$ be the group of finitely supported local gauge transformations on the ambient field algebra, and let α_g denote its action. Define

$$\mathcal{A}^G = \{A \in \mathcal{A} : \alpha_g(A) = A \text{ for every } g \in \mathcal{G}_{\text{loc}}\}. \quad (4.1)$$

The electric and plaquette terms are gauge invariant; the hopping contains its parallel transporter; the determinant projector is a $\text{Spin}(10)$ singlet; and the seam charge is neutral as assembled. Thus

$$\alpha_g(\Phi(X)) = \Phi(X), \quad \alpha_g \circ \tau_t^\Lambda = \tau_t^\Lambda \circ \alpha_g. \quad (4.2)$$

Proposition 4.1 (Compatible constrained dynamics). *The ambient embeddings (1.1) restrict to an inductive family of local gauge-invariant observables, $\tau_t(\mathcal{A}^G) = \mathcal{A}^G$, and $\tau_t|_{\mathcal{A}^G}$ is the thermodynamic dynamics of the constrained theory. For every finite region F , let P_F be the product/Haar Gauss projection. If A is gauge invariant and supported in the interior of F , then*

$$[\iota(A), P_F] = 0, \quad P_F \iota(A) P_F = \iota(A) P_F. \quad (4.3)$$

Proof. If A is fixed by every local gauge transformation, so is $A \otimes \mathbf{1}$; hence the ambient embedding preserves the fixed-point condition. Equation (4.2) implies that every $\tau_t^\Lambda(A)$, and therefore its norm limit, is gauge invariant. This proves invariance and consistency of the restricted automorphism group.

For a finite gauge group, $P_F = \prod_{x \in F} |G|^{-1} \sum_{g \in G} U_x(g)$; for a compact group, replace each sum by normalized Haar integration. Gauge invariance says $U_x(g) \iota(A) U_x(g)^* = \iota(A)$ for every factor. Thus $\iota(A)$ commutes with each averaged projector and with their product, proving (4.3). This is the standard gauge-invariant-subalgebra treatment of the local Gauss law [5, 6]. $\square$

Remark 4.2 (What is not compatible). The projected spaces $P_\Lambda \mathcal{H}_\Lambda$ do not generally embed as $P_\Lambda \mathcal{H}_\Lambda \otimes \mathcal{H}_{\Lambda' \setminus \Lambda} \subset P_{\Lambda'} \mathcal{H}_{\Lambda'}$: boundary Gauss laws correlate matter with new links and boundary flux sectors. Therefore no naive tensor embedding of physical Hilbert spaces is claimed. The honest constrained thermodynamic object is $\mathcal{A}^G$, for which (4.3) is exactly the required projector–restriction commutation. A constraint-violating operator need not commute with P_F ; the numeric mutant has commutator norm one.

5 Existence of equilibrium states

Theorem 5.1 (Ground and KMS states). *The infinite-volume system $(\mathcal{A}, \tau)$, and its invariant subalgebra $(\mathcal{A}^G, \tau|_{\mathcal{A}^G})$, have at least one ground state. For every $\beta \in (0, \infty)$, they have at least one β -KMS state.*

Proof. The state space $S(\mathcal{A})$ is weak-* compact. Choose an exhaustion $\Lambda_n \uparrow \Gamma$, a normalized ground vector of H_{Λ_n} , and extend its vector state from $\mathcal{A}_{\Lambda_n}$ to $\mathcal{A}$. Gauge-average the extension if necessary. A subnet converges weak-* to a state ω_∞ . If A is local, then for all sufficiently large n , the finite-volume ground-state variational inequality gives

$$-i \omega_n(A^* \delta_{\Lambda_n}(A)) = \omega_n(A^* [H_{\Lambda_n}, A]) \geq 0. \quad (5.1)$$

For local A , the commutator stabilizes to $\delta(A) = i \sum_{X: X \cap \text{supp } A \neq \emptyset} [\Phi(X), A]$. Passing to the weak-* limit proves the ground-state condition $-i \omega_\infty(A^* \delta(A)) \geq 0$ on the local core, hence ω_∞ is a ground state.

For finite β , take the finite Gibbs state

$$\omega_{n,\beta}(A) = \frac{\text{Tr}_{\Lambda_n}(e^{-\beta H_{\Lambda_n}} A)}{\text{Tr}_{\Lambda_n}(e^{-\beta H_{\Lambda_n}})} \quad (5.2)$$

and any extension to $\mathcal{A}$, again gauge-averaged if desired. Weak-* compactness supplies a convergent subnet ω_β . To pass the KMS identity without assuming complex-time volume convergence, smear a local A with a Gaussian f :

$$A_f = \int_{\mathbb{R}} f(t) \tau_t(A) dt, \quad A_{f,n} = \int_{\mathbb{R}} f(t) \tau_t^{\Lambda_n}(A) dt. \quad (5.3)$$

Theorem 3.1, isometry, and the Gaussian tail imply $A_{f,n} \rightarrow A_f$ in norm; the same holds after the imaginary translation of the entire Gaussian smear. The finite Gibbs KMS identity

$$\omega_{n,\beta}(A_{f,n}B) = \omega_{n,\beta}(B\tau_{i\beta}^{\Lambda_n}(A_{f,n})) \quad (5.4)$$

therefore passes to the subnet limit. Gaussian-smearred local elements are a norm-dense entire analytic subalgebra. Equation (5.4) on this core is equivalent to the full β -KMS condition, so ω_β is a KMS state. Restriction gives states on $\mathcal{A}^G$; alternatively, gauge averaging preserves both ground and KMS conditions because the gauge action commutes with τ . $\square$

6 Numeric twin

The companion

`experiments/tfpt-discovery/lieb_robinson_witness_probe.py`

uses the exact nearest-neighbour physical-edge sector of the assembled model while comparing it against the complete class constant (2.7). This is the same finite cone protocol as the earlier IR-universality witness. It reports:

quantity	value
v_{LR} (proved class bound)	387.639069990831
v_{measured} (commutator threshold 0.02)	2.000000000000
outside-cone leakage	8.238×10^{-3}

(6.1)

For the one-site projector, $t = 10^{-3}$, and nested open volumes, the specialization of (3.4) gives

b	$\ \tau_t^{\Lambda_b}(A) - \tau_t^{\Lambda_{64}}(A)\ $	$S_2(b-2)(e^{2\epsilon kt} - 1)$
8	0	$9.246214927512 \times 10^{-1}$
16	0	$3.279645827421 \times 10^{-2}$
32	0	$2.164381663985 \times 10^{-5}$

(6.2)

The zeros are numerical precision statements, not analytic finite-speed claims. A $1/r$ all-to-all mutant, which is outside (2.1), gives $1.048309991863 \times 10^{-4}$ at $b = 32$, violating the inapplicable finite-range line by a factor 4.843462. The Gauss-positive control has $\|[A, P_G]\| = \|P_G A P_G - A P_G\| = 0$; the constraint-violating embedding has $\|[A_{\text{bad}}, P_G]\| = 1$. All eight checks pass.

7 Combined theorem and exact boundary

Theorem 7.1 (Mandatory thermodynamic-dynamics leg). *For the TFPT master-Hamiltonian class (1.4) with the frozen finite-dimensional local factors and coefficients of Table 1:*

- (i) *the finite-volume dynamics satisfy the uniform Lieb–Robinson bound (2.2) with the explicit estimate $v_{\text{LR}} = 32086e/225$;*
- (ii) *the thermodynamic-limit dynamics exist in norm on all local observables and define a strongly continuous automorphism group;*
- (iii) *the group restricts consistently to the Gauss-invariant quasilocal algebra, with projector compatibility in the precise form (4.3); and*
- (iv) *at least one thermodynamic ground state and at least one β -KMS state for every $\beta > 0$ exist.*

No continuum limit is used in any part of the proof.

Proof. Part (i) is Theorem 2.1 together with (1.6). Part (ii) is Theorem 3.1, including the explicit norm-Cauchy estimate (3.4). Part (iii) is Proposition 4.1; the warning following it excludes the false physical-Hilbert-space embedding. Part (iv) is Theorem 5.1. $\square$

BOUNDARY BOX — what this does not give.

This theorem does *not* give uniqueness of the ground or KMS state; phase structure and spontaneous symmetry breaking remain undecided. It does *not* prove that the finite-volume spectral gap survives the thermodynamic limit. It does *not* establish an interacting continuum/scaling limit, Lorentz invariance, a common infrared light cone, or IR universality. Those are precisely the remaining T3/T5 contents. The result proved here is the mandatory dynamics leg only: a uniformly local Hamiltonian class, a volume-independent Lieb–Robinson estimate, norm thermodynamic-limit dynamics, a compatible gauge-invariant quasilocal algebra, and existence (not uniqueness) of equilibrium states.

Conclusion

The amended contract makes the continuum limit optional but the thermodynamic limit mandatory. The latter is now proved at the Hamiltonian-class level for the concrete finite-group, finite-local-space TFPT assembly. The proof is insensitive to finite-volume boundary shape and does not infer thermodynamics from a finite scan. Its explicit velocity is conservative, while the numerical witness checks the expected cone, the norm-convergence inequality, and both destructive mutants. The boundary box prevents this dynamics theorem from being misread as a gap, phase-uniqueness, or IR-universality theorem.

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